

Canterbury
Christ Church
University

B.Eng. (Hons) in Software Engineering

Software Engineering Cohort 4 – Trimester 1

U19972 – Advanced Operating Systems

Assessment 1

Module Component	50%
Assessment Submission	31 st March 2026
Expected Feedback Date	15 Working Days
Where to Submit	Blackboard/Turnitin
What to Submit	2000-word single report of a high professional standard.

Assessment Instructions

This assessment evaluates your ability to apply operating system management techniques and automation tools in a real-world context. It is structured into three scripting-based tasks requiring you to write Bash or Python scripts to manage processes, schedule jobs, validate files, and implement automated system controls.

1. University Data Centre Process and Resource Management System (Bash Scripting Only)

This task aligns with Learning Outcome 2 and demonstrates system automation techniques for process monitoring, disk management, and log control, which are key tools for operating system administration.

The University Data Centre hosts multiple research and academic services on shared Linux servers. Performance degradation has been observed due to unmanaged background processes, excessive memory consumption, and uncontrolled growth of log files.

Your task is to write a Bash script that simulates an intelligent system administration tool capable of executing the following operations via a menu driven interface.

Requirements

- a. Process Monitoring and Management

- Display current CPU and memory usage.
 - List the top ten memory consuming processes including PID, user, CPU percentage and memory percentage.
 - Terminate a selected process only after confirmation
 - Prevent termination of critical system processes.
- b. Disk Inspection and log archiving
- Inspect disk usage of a specified directory.
 - Detect log files larger than 50MB
 - Create an ArchiveLogs directory if it does not exist.
 - Compress large log files into ArchiveLogs with a timestamped filename.
 - Display a warning if ArchiveLogs exceeds 1GB.
- c. Logging System
- Maintain a system_monitor_log.txt file.
 - Record all administrative actions with timestamps.
- d. Exit Mechanism
- Implement a Bye exit command with a Y or N confirmation prompt.

2. University High Performance Computing Job Scheduler (Bash / Python)

This task aligns with Learning Outcome 4 which requires using Bash or Python scripting to implement scheduling algorithms and manage queue systems efficiently within a POSIX compliant environment. The University High Performance Computing Laboratory provides limited graphical processing resources to research students. Manual allocation has resulted in unfair scheduling and inefficient resource usage.

You are required to develop a script that manages computational job requests and processes them using Round Robin or Priority Scheduling.

Requirements

- a. User Menu
- View pending jobs
 - Submit a job request
 - Process job queue using either round robin or priority scheduling
 - View completed jobs

- Exit system with confirmation
- b. Queue based Scheduling System
 - Each job must include student ID, job name, estimated execution time and priority value between 1 to 10.
 - Round robin must use a time quantum of five seconds
 - Priority scheduling must execute highest priority jobs first.
- c. Logging System
 - Log every job submission and execution event in scheduler_log.txt
 - Record student ID, job name, scheduling type and timestamp.
- d. Data Storage
 - Store pending jobs in job_queue.txt
 - Store completed jobs in completed_jobs.txt

3. Secure Examination Submission and Access Control System (Bash & Python)

This task aligns with Learning Outcomes 2 and 4. It demonstrates system administration techniques for file validation, duplicate detection, logging mechanisms, and security monitoring.

The Examination Board requires a secure submission system that validates file formats, prevents duplicate submissions, and detects suspicious login behaviour.

You must develop a script that validates student submissions and monitors login attempts.

Requirements

- a. Menu System
 - Submit an assignment
 - Check if a file has already been submitted.
 - List all submitted assignments
 - Simulate login attempt
 - Exit system with confirmation.
- b. File validation
 - Only accept .pdf and .docx files
 - Reject files larger than 5MB.
- c. Duplicate Submission Detection
 - Reject submissions with identical filename and content.

- Log all submissions in submission_log.txt
- d. Access control and suspicious activity detection.
 - Lock accounts after three failed login attempts
 - Detect repeated login attempts within sixty seconds
 - Log all login attempts with timestamps.

Report Writeup (800 Words)

Alongside your source code, you must submit an 800-word report that explains your implementation in detail.

The report should,

1. Explain your design choices and justify command selection.
2. Connect your implementation to advanced operating system concepts such as process management, scheduling algorithms, file system operations, and automation techniques.
3. Include at least three academic or technical references using Harvard referencing.
4. Identify challenges encountered and how they were resolved.
5. Provide execution steps and expected output with screenshots.

Marking Scheme

- Task 1 – Process and Resource Management Systems – 30 Marks
- Task 2 – High Performance Computing Scheduler – 30 Marks
- Task 3 – Secure Submission and Access Control System – 30 Marks
- Report Writeup – 10 Marks

Assessment Rubric

Each task is graded out of 30 marks while report is worth 10 marks, making a total of 100 marks.

The final grade is determined by the sum of all tasks and the report.

Task 1

University Data Centre Process and Resource Management System (30 Marks)

Criteria	Marks	Description
Menu system and user Interaction	6	The script provides a complete interactive menu. All required administrative options are available. Invalid inputs are handled appropriately without system failure.
Process monitoring and management	8	Processes are displayed correctly including PID, user, CPU percentage, and memory percentage. Safe termination is implemented with confirmation. Critical processes are protected from termination.
Disk Inspection and Log Archiving	8	Large log files are detected correctly. ArchiveLogs directory is created if required. Files are compressed with timestamps. Warning is displayed when storage threshold exceeds 1GB.
Logging system	4	All administrative actions are recorded in system_monitor_log.txt with timestamps and clear descriptions.
Exit Mechanism and Confirmation	4	A Bye command is implemented and requires Y or N confirmation before termination.

Task 2

University High Performance Computing Job Scheduler (30 Marks)

Criteria	Marks	Description
Menu system and user interaction	6	The system provides complete navigation options including job submission, queue viewing, execution, and exit. Input validation is implemented correctly.
Scheduling algorithm implementation	8	Round Robin scheduling operates correctly using a five second time quantum. Priority scheduling correctly executes highest priority jobs first. Execution order is clearly displayed.
Queue and data storage system	8	Jobs are stored correctly in job_queue.txt. Completed jobs are recorded in completed_jobs.txt. Queue ordering is accurate for both scheduling methods.
Logging System	4	All scheduling events are recorded in scheduler_log.txt including timestamp, student

		ID, job name, scheduling type, and execution result.
Exit mechanism and confirmation	4	Bye command is implemented and requires confirmation before exiting the system.

Task 3

Secure examination submission and access control system (30 Marks)

Criteria	Marks	Description
Menu system and user interaction	6	The system provides a complete menu and handles invalid inputs appropriately.
File validation	8	Only pdf and docx files are accepted. Files larger than 5MB are rejected. Invalid formats are identified immediately.
Duplicate detection and access control	8	The system detects duplicate filenames and identical content. Duplicate submissions are rejected. Accounts are locked after three failed login attempts. Suspicious login patterns are detected.
Logging System	4	All submission and login events are recorded in submission_log.txt including timestamp, student ID, filename, and status
Exit mechanism and confirmation	4	A Bye command is implemented with Y or N confirmation before termination.

Report Write Up (10 Marks)

Criteria	Marks	Description
Code explanation and screenshots	4	The report clearly explains implementation decisions and includes evidence of script execution.
Challenges and solutions	2	Technical challenges are identified and solutions are explained clearly.
Execution steps and expected output	2	Clear step by step execution instructions are provided along with expected behaviour.
Professional Formatting and Referencing	2	The report is well structured, written in academic language, and follows Harvard referencing style.

Additional Submission Requirements

GITHUB Repository

You must maintain a private GitHub repository for this assessment. Your repository must contain

- All source code files
- Clear commit history showing development progression
- A readme file explaining how to execute each script
- You must include the GitHub repository link inside your written report.
- Failure to provide a repository link may result in mark reduction.

20 minute live demonstration (recordings will be kept for documentation)

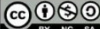
You must make a live demonstration of your current work via MS Teams. Failure to demo the artifact may result in mark reduction.

Generative Artificial Intelligence (GenAI) and Assessments

Before approaching any assessment you should read the student guidance on the use of GenAI: [Welcome to your generative AI guidance](#). This will ensure you are aware of the ethical, legal and learning aspects of using GenAI for your studies.

Each module's assessments will require a different approach to the use of Generative Artificial Intelligence (GenAI). Your Module Lead has evaluated this module's assessment(s) using the Artificial Intelligence Assessment Scale (AIAS). You can learn more about the AIAS at aiassessmentscale.com.

If you would like to understand more about how CCCU's staff understand the use of GenAI in your learning, you can read the [Staff Guidance](#).

1	No AI	The assessment is completed entirely without AI assistance in a controlled environment, ensuring that students rely solely on their existing knowledge, understanding, and skills. You must not use AI at any point during the assessment. You must demonstrate your core skills and knowledge.
2	AI Planning	AI may be used for pre-task activities such as brainstorming, outlining and initial research. This level focuses on the effective use of AI for planning, synthesis, and ideation, but assessments should emphasise the ability to develop and refine these ideas independently. You may use AI for planning, idea development, and research. Your final submission should show how you have developed and refined these ideas.
3	AI Collaboration	AI may be used to help complete the task, including idea generation, drafting, feedback, and refinement. Students should critically evaluate and modify the AI suggested outputs, demonstrating their understanding. You may use AI to assist with specific tasks such as drafting text, refining, and evaluating your work. You must critically evaluate and modify any AI-generated content you use.
4	Full AI	AI may be used to complete any elements of the task, with students directing AI to achieve the assessment goals. Assessments at this level may also require engagement AI to achieve goals and solve problems. You may use AI extensively throughout your work either as you wish, or as specifically directed in your assessment. Focus on directing AI to achieve your goals while demonstrating your critical thinking.
5	AI Exploration	AI is used creatively to enhance problem-solving, generate novel insights, or develop innovative solutions to solve problems. Students and educators co-design assessments to explore unique AI applications within the field of study. You should use AI creatively to solve the task, potentially co-designing new approaches with your instructor.
 Perkins, Furze, Roe & McVaugh (2024). The AI Assessment Scale		

Your Module Lead has evaluated this module's assessments as requiring the following level of interaction with GenAI:

Assessment 1: AI Planning

Assessment 2: AI Planning

At CCCU all assessments require a declaration regarding the level of GenAI use.

Under CCCU's Academic Integrity Policy unacknowledged inclusion of GenAI is considered academic misconduct. See the [Academic Integrity and Misconduct](#) webpages for more information.

Please select the most appropriate statement from the choice below and insert it at the start of your work:

- No GenAI was used in the preparation, planning or creation of this work. [AIAS Level 1]
- I acknowledge the use of outputs from [insert the name of generative AI tool(s) used] in the learning, preparation, planning or proofreading of this work. [AIAS Level 2]
- I acknowledge collaboration with [insert the name of generative AI tool(s) used] in this work, and the inclusion of outputs in modified form. [AIAS Level 3]
- I acknowledge [insert the name of generative AI tool(s) used] as partner(s) in the creation of this work. [AIAS Levels 4 & 5]

Referencing GenAI

If your assessment is rated at levels 2, 3 or 4, **you are required to reference where you have used GenAI outputs in the body of your work.** For guidance on how to reference GenAI in your work see Cite Them Right 13th Edn, or later. You can use the textbook or visit the [Cite Them Right](#) website – both are accessible via [LibrarySearch](#) (for the web version, login using your CCCU email). You can speak to your module tutor, module lead, subject Librarian or Learning Developer for more guidance on the use of GenAI in your learning and assessment.

Resources:

Some resources to get started with research are provided below:

1. <https://www.hillingdontimes.co.uk/news/national/25455598.jaguar-land-rover-says-data-accessed-cyber-attack/>
2. <https://www.telegraph.co.uk/business/2025/09/10/jaguar-land-rover-admits-data-stolen-in-cyber-attack/>
3. <https://gbhackers.com/jaguar-land-rover-confirms-cyberattack/>
4. <https://www.caranddriver.com/news/a66125275/jlr-cyberattack-timeline/>

Learning Outcomes Assessed

LO2

Exploit tools and techniques that allow administrators to manage the operating system efficiently and critically evaluate their efficacy.

LO4

Demonstrate the ability to competently operate the UNIX command line environment including knowledge of the POSIX standard and advanced shell scripting.